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SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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REVIEW 2-PRESENTATION

Predictive Wildfire Spread Simulator with Evacuation Route Optimizer

CSA0915– Java Programming

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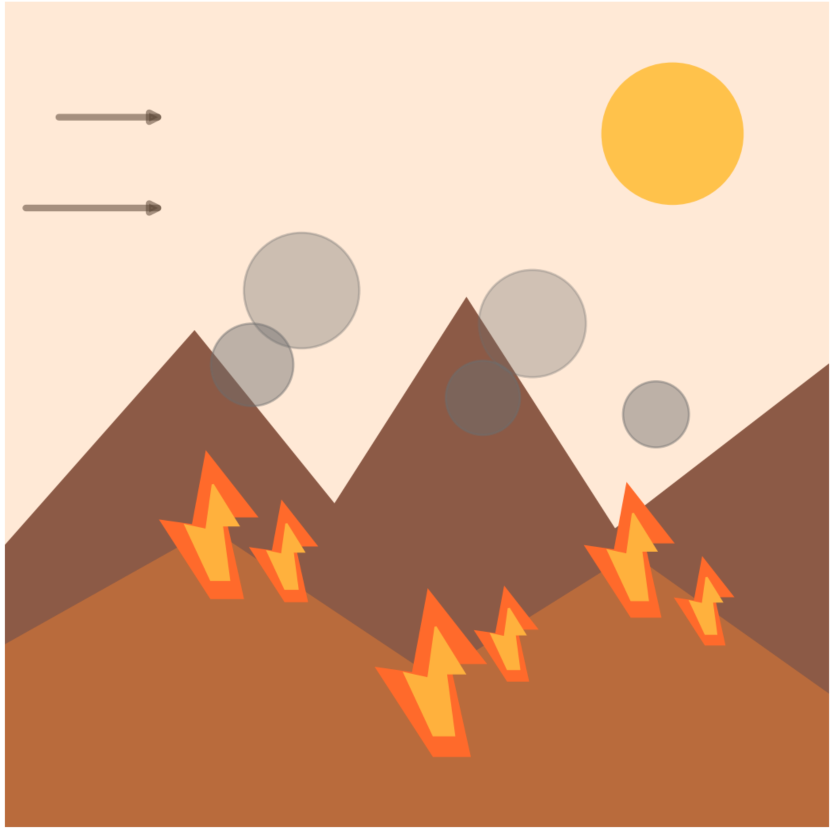
INTRODUCTION

- Wildfires are among the most destructive natural disasters, causing loss of life, forests, and property.
- Climate change, high temperatures, strong winds, and dry vegetation have increased wildfire frequency and intensity.
- Early prediction of wildfire spread is essential to reduce damage and improve emergency response.
- FireWatch is an intelligent system that predicts the direction and speed of wildfire spread using environmental data.
- The system considers factors such as wind speed, temperature, humidity, vegetation, and terrain.



PROBLEM STATEMENT

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- Wildfires spread rapidly due to changing weather conditions, making them difficult to predict.
- Existing wildfire monitoring systems mainly detect fires after they have already started, limiting response time.
- Delayed prediction and evacuation can lead to significant loss of lives, property, forests, and wildlife.
- Emergency responders often face challenges in identifying the safest and fastest evacuation routes.
- Real-time environmental factors such as wind speed, temperature, humidity, and terrain are not always integrated into decision-making.

PROPOSED SYSTEM

- To develop an intelligent system for predicting wildfire spread.
- To analyze wind, temperature, humidity, vegetation, and terrain for better accuracy.
- To simulate wildfire spread direction and speed in real time.
- To identify the safest and shortest evacuation routes.
- To provide early warnings for emergency response teams.
- To reduce loss of life, property, and forests through timely evacuation.
- To improve disaster management via combined prediction and route planning.

MODULES

Module 1: Data Collection & Wildfire Prediction

Module 2: Wildfire Spread Simulation

Module 3: Evacuation Route Optimization & Alert System

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SYSTEM ARCHITECTURE

• FireWatch combines environmental data analysis, wildfire prediction, spread simulation, evacuation route optimization, and alert generation.

• Input data includes weather conditions, vegetation, terrain, road information, safe locations, and the detected fire location.

• Each stage produces information that becomes the input for the next stage.

Environmental Data
↓
Prediction Engine
↓
Spread Simulation
↓
Risk Map
↓
Route Optimizer
↓
Alerts

MODULE 1: DATA COLLECTION & WILDFIRE PREDICTION

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- Collect temperature, humidity, wind speed and direction, vegetation condition, terrain slope, and current fire-location data.
- Preprocess the data by checking missing values and converting all inputs into a suitable format.
- Analyze the environmental conditions to estimate the likely direction and relative speed of wildfire spread.
- Output: predicted spread parameters and an initial risk estimate.

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MODULE 1 – WORKING

- Step 1: Receive environmental and geographic inputs.
- Step 2: Validate and preprocess the collected data.
- Step 3: Analyze factors influencing fire behaviour.
- Step 4: Predict the likely spread direction and speed.
- Step 5: Pass the prediction to the simulation module.

Input	Process	Analysis	Output	Update

MODULE 2: WILDFIRE SPREAD SIMULATION

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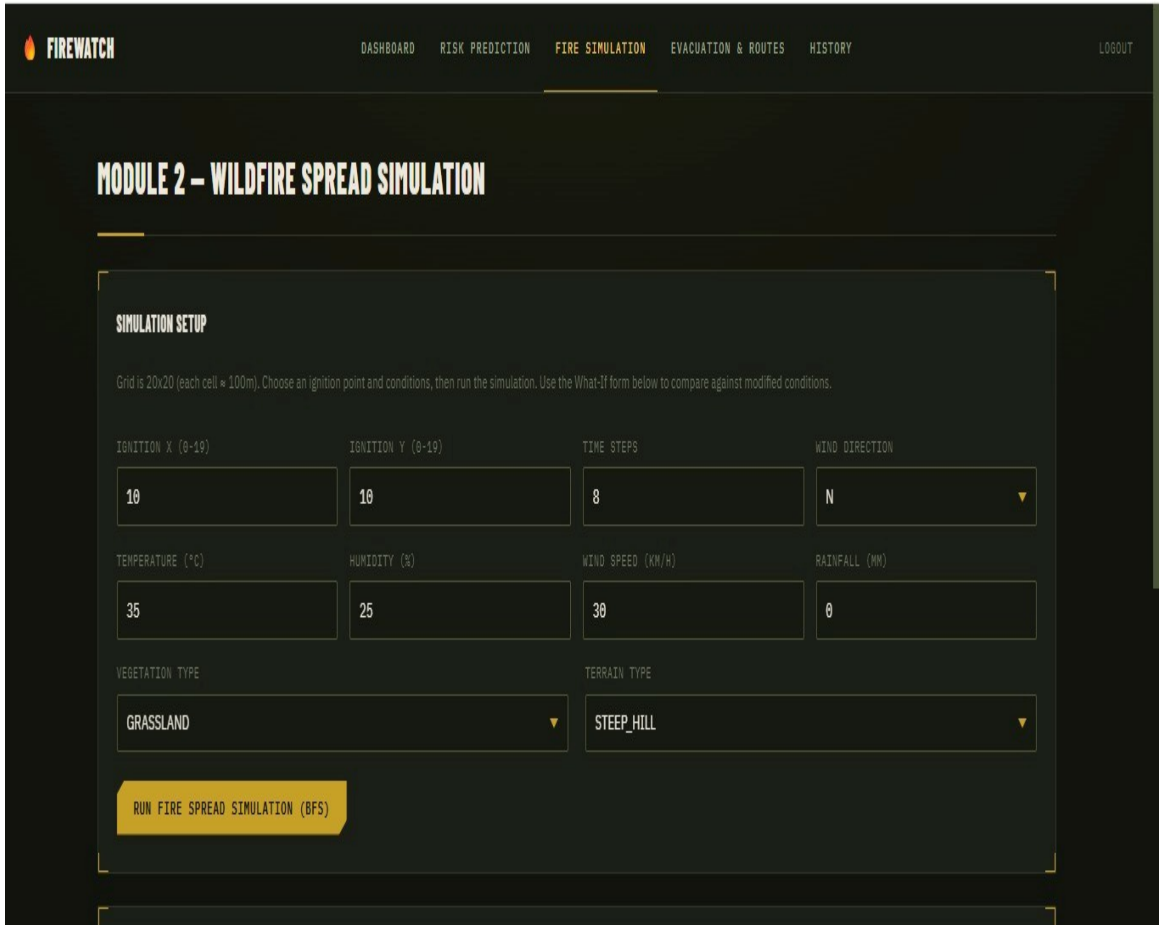
- Represent the affected area using map cells or geographic zones.
- Start the simulation from the detected wildfire location.
- Update neighbouring zones according to wind, humidity, vegetation, terrain, and predicted spread behaviour.
- Generate a time-based view showing how the risk area may expand.
- Output: a dynamic wildfire risk map.



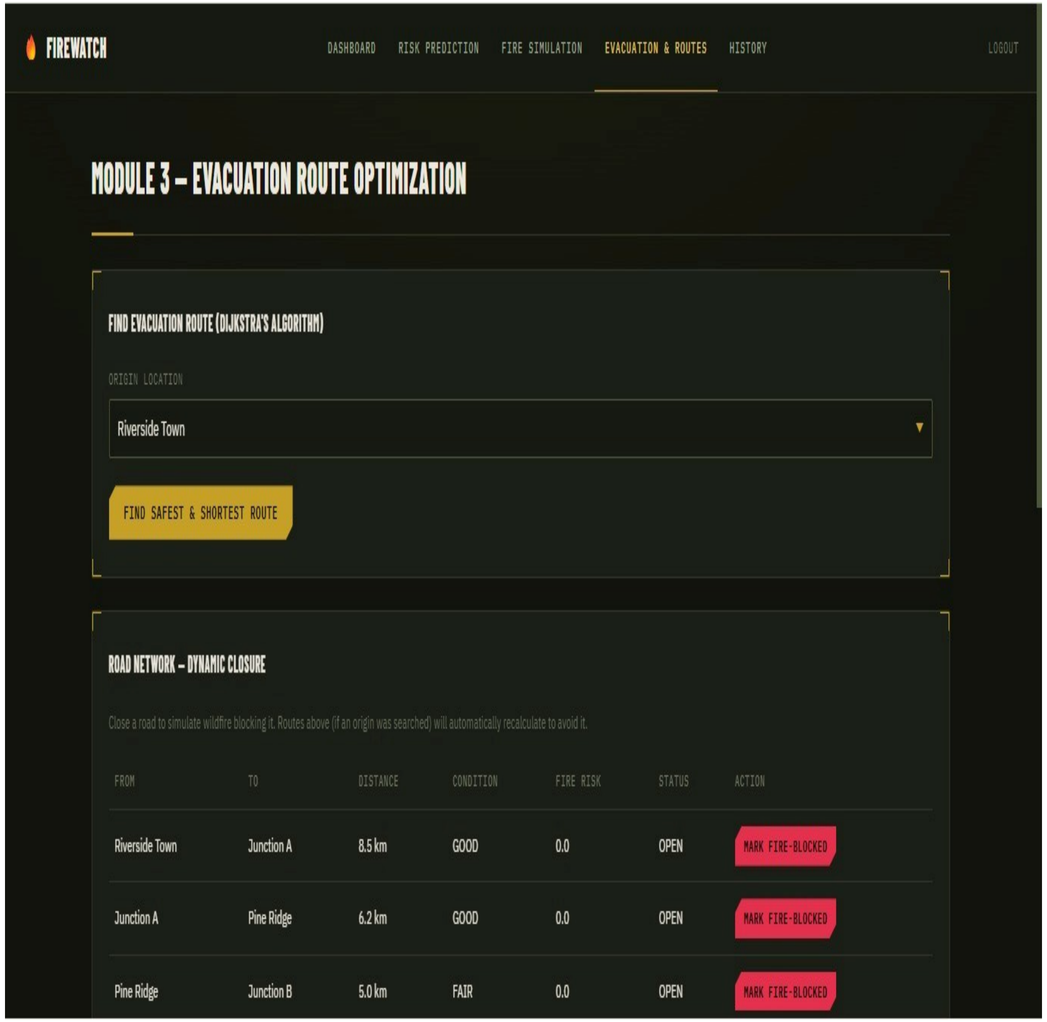
MODULE 2 – WORKING

- Step 1: Initialize the map and fire source location.
- Step 2: Divide the region into zones or cells.
- Step 3: Calculate the influence of environmental conditions.
- Step 4: Update the predicted fire boundary for each time interval.
- Step 5: Mark zones according to their risk level.

Input | **Process** | **Analysis** | **Output** | **Update**

MODULE 3: EVACUATION ROUTE OPTIMIZATION & ALERT SYSTEM

- Combine the wildfire risk map with roads, exits, and safe shelter locations.
- Avoid or assign high cost to routes passing through predicted high-risk zones.
- Use graph-based algorithms such as Dijkstra's or A* to identify safer routes.
- Recalculate the route when the predicted wildfire situation changes.
- Generate alerts containing risk status and recommended evacuation information.



MODULE 3 – WORKING

• Step 1: Receive the latest risk map from the simulation module.

• Step 2: Identify the user's or affected area's starting point.

• Step 3: Evaluate available roads and remove unsafe paths.

• Step 4: Calculate a safer route to a safe destination.

• Step 5: Display the route and generate an alert.

Input | **Process** | **Analysis** | **Output** | **Update**

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METHODOLOGY

- Data collection provides the environmental and geographic information.
- The prediction engine estimates the possible fire movement.
- The simulation module converts the prediction into a changing risk map.
- The route optimizer searches for a safer path using the current risk information.
- The alert system communicates the recommendation and updates it when new conditions are available.

Collect | **Predict** | **Simulate** | **Risk Map** | **Optimize** | **Alert**

JAVA PROGRAMMING IMPLEMENTATION

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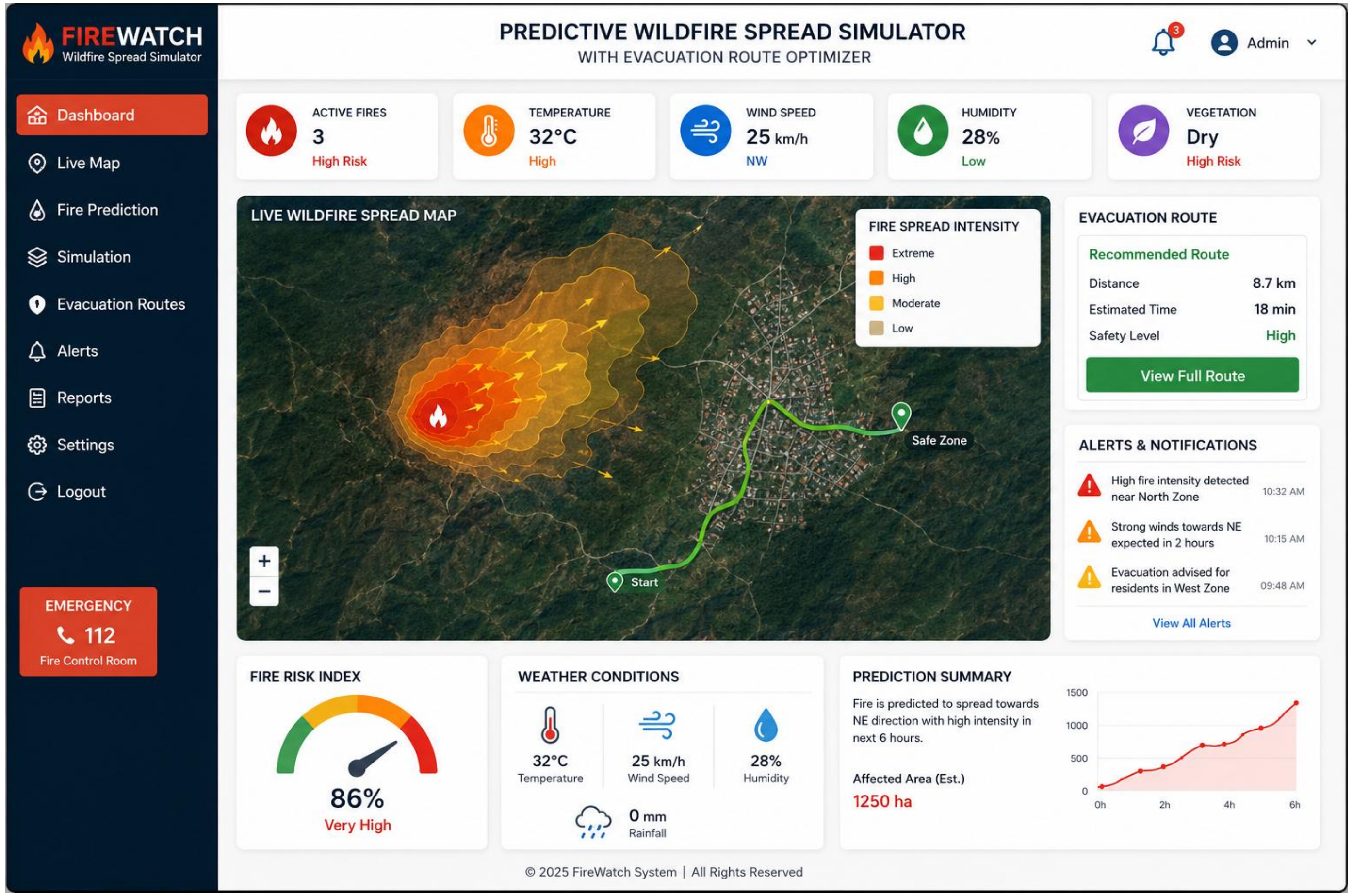
- Java can implement the project using object-oriented programming and modular classes.
- Suggested classes: WeatherData, FireCell, FirePredictor, SpreadSimulator, RiskMap, Graph, RouteOptimizer, and AlertService.
- Arrays or ArrayLists can represent simulation data and geographic zones.
- Graph data structures can represent roads and locations.
- Dijkstra's or A* algorithm can be used for evacuation route calculation.



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REAL-LIFE EXAMPLE SCENARIO

- Assume a wildfire is detected near a forest boundary close to a residential area.
- Strong wind and low humidity indicate that the fire may spread more rapidly in a particular direction.
- The simulator marks the predicted path as a higher-risk zone.
- The route optimizer avoids roads entering the high-risk area and searches for an alternative route.
- Emergency teams can use the result to support evacuation decisions.

Forest 🔥 → Risk Zone
↘
Alternative Route →
🏠

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EXPECTED OUTPUTS

- Predicted wildfire spread direction and relative speed.
- Dynamic wildfire risk map.
- Identification of areas that may become unsafe.
- Recommended evacuation route based on current predicted risk.
- Alert messages for emergency response and evacuation.
- Updated recommendations when input conditions change.

📍 Risk Map

🧭 Route

🔔 Alert

ADVANTAGES

- Combines prediction, simulation, route planning, and alerts in one system.
- Provides earlier decision-support information for emergency situations.
- Supports route selection using predicted wildfire risk.
- Modular design makes testing and future expansion easier.
- Demonstrates a real-world application of Java programming and graph algorithms.

LIMITATIONS

- Prediction quality depends on the accuracy and availability of input data.
- Actual wildfire behaviour can change rapidly because environmental conditions are dynamic.
- Real-world deployment requires reliable maps, weather feeds, and continuous updates.
- Recommended routes must be validated with real road and emergency information.
- The current academic prototype should be tested extensively before operational use.

⚠

Data Quality
Real-time Changes
Validation Needed

FUTURE ENHANCEMENTS

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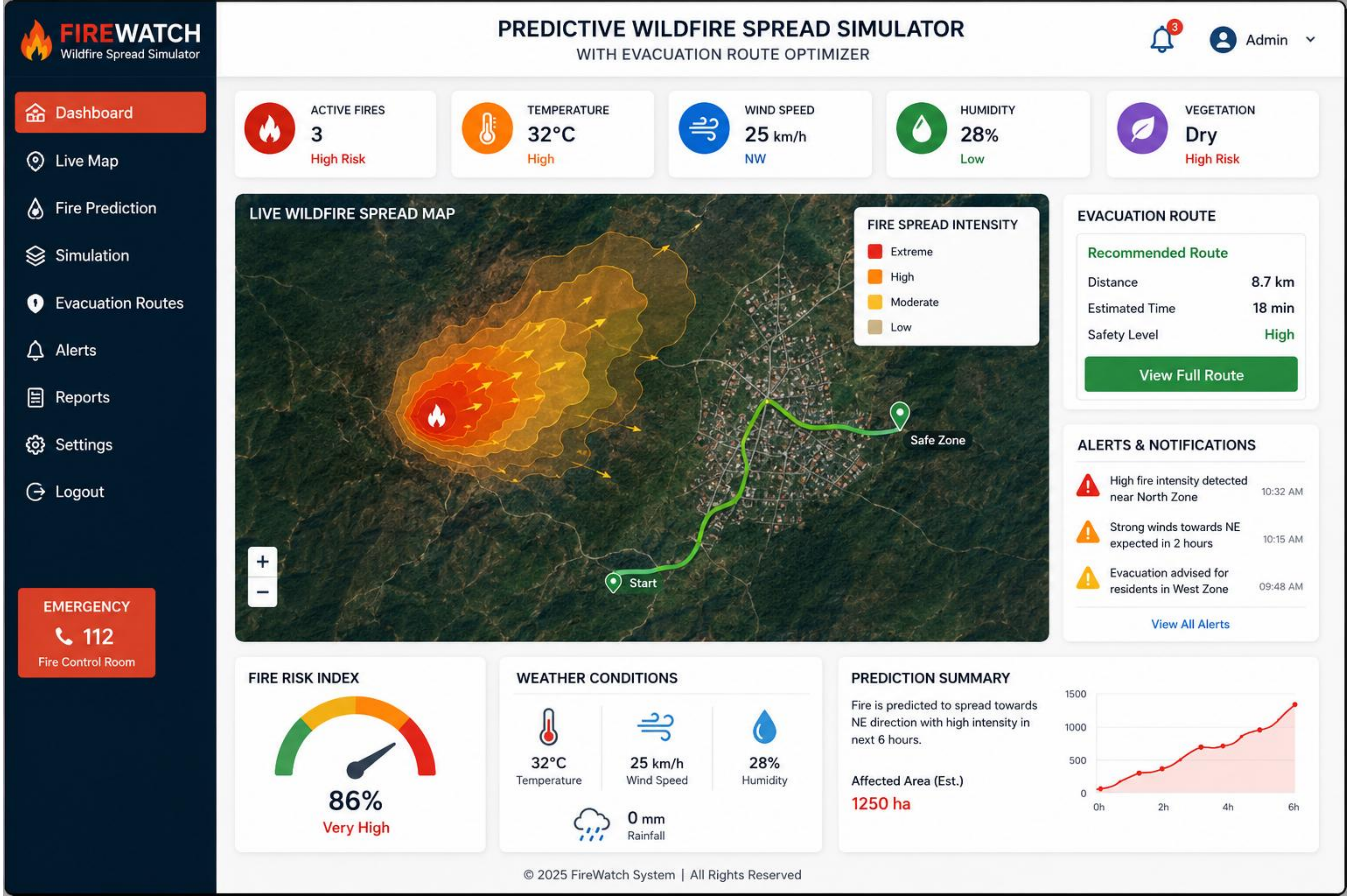
- Integrate real-time weather data and sensor or satellite observations.
- Use machine learning models trained on historical wildfire information.
- Add GIS maps with live road closures and safe shelter details.
- Develop a mobile application for location-aware guidance.
- Include traffic conditions and multiple evacuation-route options.
- Create an emergency authority dashboard for monitoring the situation.

AI + GIS
Real-time Data
Mobile App

CONCLUSION

- FireWatch integrates wildfire prediction, spread simulation, and evacuation route optimization into a single workflow.
- The three modules provide a clear structure for developing the system.
- Environmental information is converted into a predicted risk map that supports route planning.
- The project provides a practical Java-based application involving object-oriented programming, data structures, and graph algorithms.


FIREWATCH
Wildfire Spread Simulator
PREDICTIVE WILDFIRE SPREAD SIMULATOR
WITH EVACUATION ROUTE OPTIMIZER
3
Admin
Dashboard
Live Map
Fire Prediction
Simulation
Evacuation Routes
Alerts
Reports
Settings
Logout
EMERGENCY
112
Fire Control Room
ACTIVE FIRES
3
High Risk
TEMPERATURE
32°C
High
WIND SPEED
25 km/h
NW
HUMIDITY
28%
Low
VEGETATION
Dry
High Risk
LIVE WILDFIRE SPREAD MAP
FIRE SPREAD INTENSITY
Extreme
High
Moderate
Low
Safe Zone
Start
+
−
EVACUATION ROUTE
Recommended Route
Distance 8.7 km
Estimated Time 18 min
Safety Level High
View Full Route
ALERTS & NOTIFICATIONS
High fire intensity detected near North Zone 10:32 AM
Strong winds towards NE expected in 2 hours 10:15 AM
Evacuation advised for residents in West Zone 09:48 AM
View All Alerts
FIRE RISK INDEX
86%
Very High
WEATHER CONDITIONS
32°C
Temperature
25 km/h
Wind Speed
28%
Humidity
0 mm
Rainfall
PREDICTION SUMMARY
Fire is predicted to spread towards NE direction with high intensity in next 6 hours.
Affected Area (Est.)
1250 ha
1500
1000
500
0
0h
2h
4h
6h
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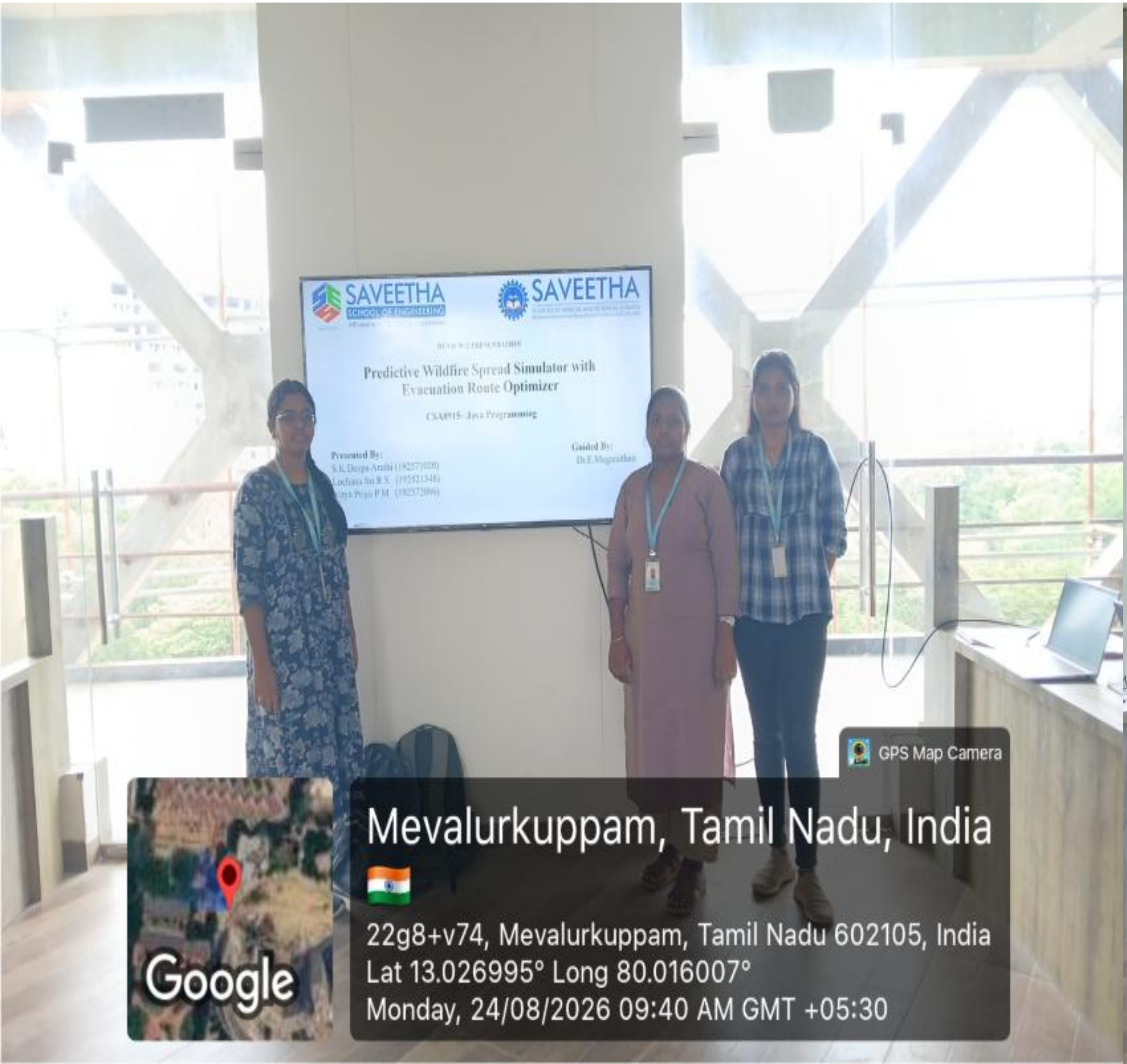
GEO TAG PHOTO



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